

Which parts of the previous QAE result survive one-factor changes?

A quantum autoencoder (QAE) robustness study — a follow-up to last meeting's architecture-search result

Baseline finding: physics-informed language-model proposals beat random and greedy circuit search by a wide margin; a closed feedback loop added a small extra gain.

Four factors, one at a time: evaluation budget · qubit count · Hamiltonian · language model.

This is a tested one-factor-slice study, not a complete factorial grid.

Tomoya Hatanaka

ML4SCI / Google Summer of Code 2026 · September 4, 2026

1 • PREVIOUS BASELINE

Where we left off: the previous baseline result

Quantum autoencoder (QAE): compress ground states into fewer qubits · 4 qubits · Ising chain · $B = 8$ · High model tier · 12 paired seeds

Input $|\psi(h)\rangle$

ground state of the 4-qubit Ising chain
 $H = -\sum Z_i Z_{i+1} - h \sum X_i$, $h \in [0.2, 2.0]$

Encoder $U\theta$

the circuit we search
(16 gates, angles trained)

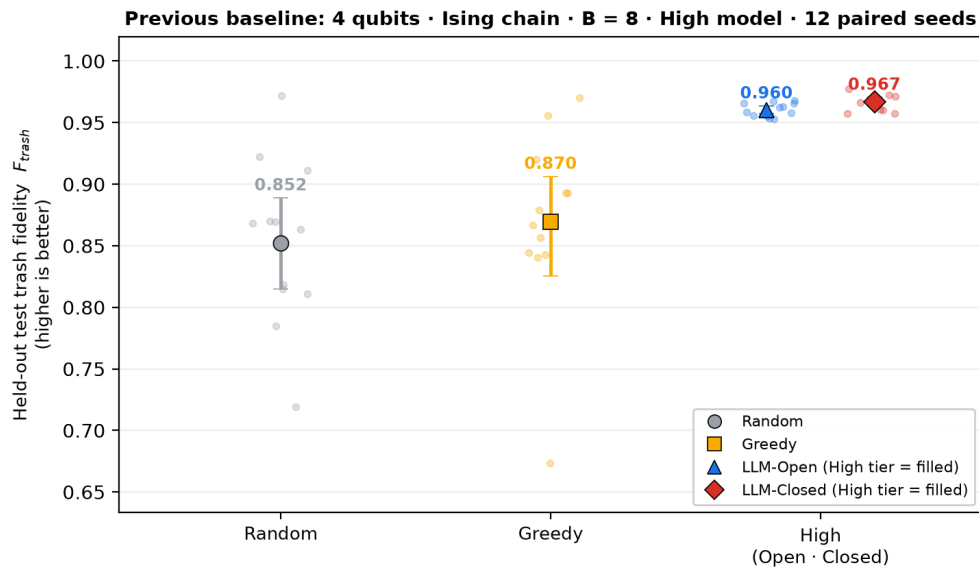
Latent qubits q_0, q_1

the 2 qubits we keep

Trash qubits q_2, q_3

should end in $|00\rangle$ for every input

Trash fidelity $F_{\text{trash}} = P(\text{trash qubits } q_2 q_3 = 00)$ — **higher is better** · measured on a held-out test set of 64 unseen h values, read once after selection



Four search methods (budget B = candidate circuits trained and evaluated per seed)

Random B uniformly sampled circuits

Greedy $B/2$ random starts + $B/2$ single-change refinements of the best

LLM-Open open-loop: all semantic proposals before any score is seen

LLM-Closed closed-loop: semantic starts + free redesigns of the current best, guided by validation scores

Two claims to stress-test

1 Semantic proposals beat non-semantic search: Open – Random +0.108 (11/12, $p < 0.001$)

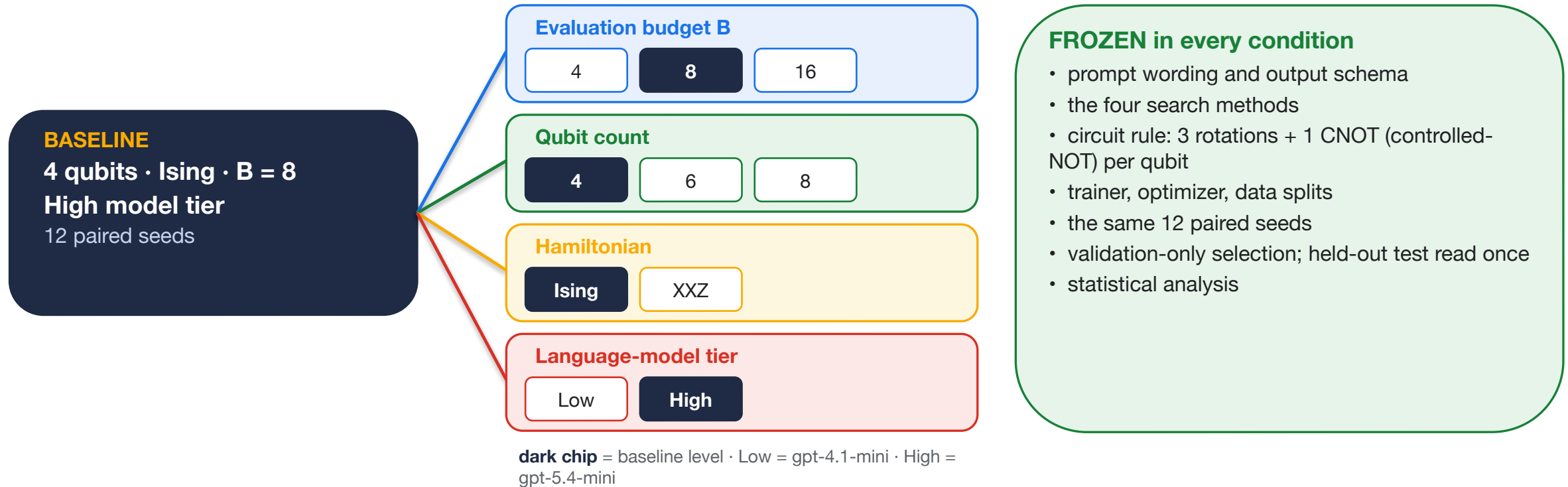
2 Closed-loop redesign adds a small extra gain: Closed – Open +0.007 (11/12, $p = 0.012$)

Next: do these two claims survive when exactly one condition changes?

Dots = the 12 paired seeds (12 independent draws of training/validation data shared by all methods); marker = mean; bar = bootstrap 95% confidence interval (CI); p = exact paired Wilcoxon test. High tier = gpt-5.4-mini.

One factor changes; everything else remains fixed

Four controlled changes leave the baseline; each moves exactly one declared factor and holds the other three



No interaction condition was tested. Each change returns to the baseline before the next factor moves — e.g. a large budget at 8 qubits was never run. XXZ = the second Hamiltonian family, an XXZ Heisenberg chain $H = \sum (X_i X_{i+1} + Y_i Y_{i+1} + \Delta Z_i Z_{i+1})$, $\Delta \in [0.2, 2.0]$, declared before the run.

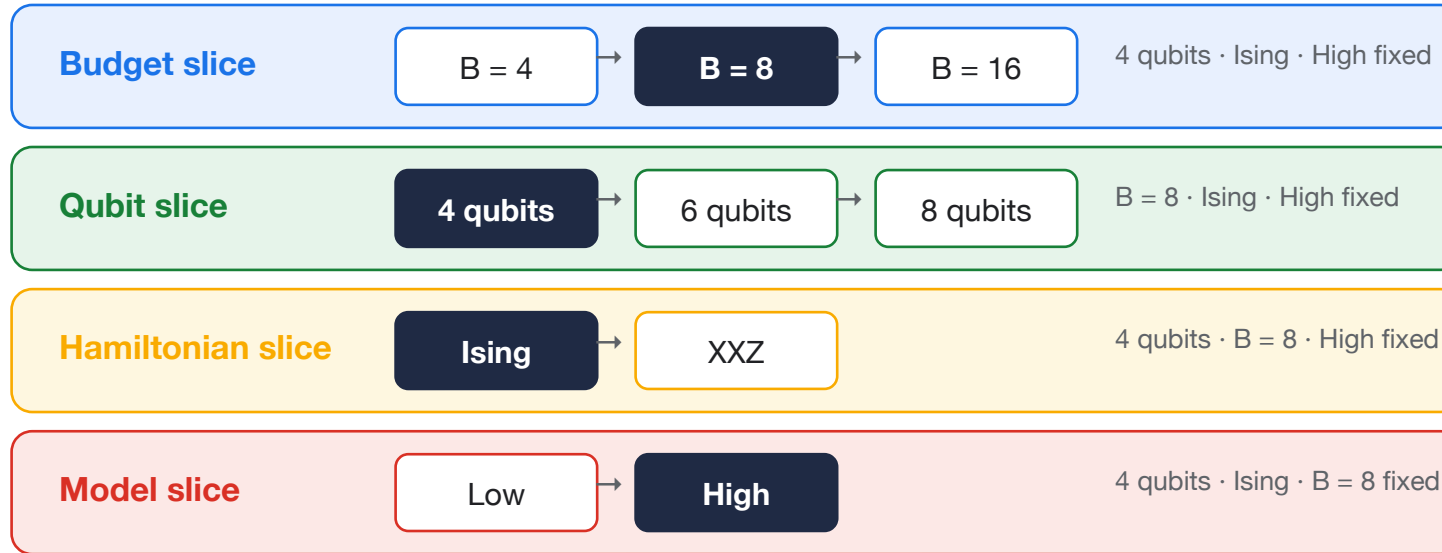
Next: because only one factor moves at a time, read the results as four tested slices, not as a full grid.

An automated check refuses any condition whose frozen components differ from the baseline's or that moves more than one factor; the baseline cell is reused, not re-run.

3 · WHAT WAS ACTUALLY TESTED, AND HOW TO READ IT

Four tested slices, one baseline, no empty panels

Every result slide shows one of these four rows; untested combinations are omitted rather than displayed as empty panels



dark chip = the baseline level, present in every slice.

Low was run only in the model slice. The other slices show Random, Greedy and High only; no Low value is invented.

Untested combinations are omitted rather than displayed as empty panels.

Model tiers (defined here once)

Low = gpt-4.1-mini **High** = gpt-5.4-mini

Fixed capability-tier labels, not the observed score.
Open / Closed name the search workflow; Low / High name the model.

Reading every result figure

- dots = 12 paired seeds · marker = mean · bar = 95% CI
- higher trash fidelity is better
- groups always: Random → Greedy → Low → High
- LLM-Open = blue triangle, LLM-Closed = red diamond
- High = filled marker, Low = hollow marker
- thick frame / shaded band = the baseline

Next: the first slice changes the amount of search — the evaluation budget B.

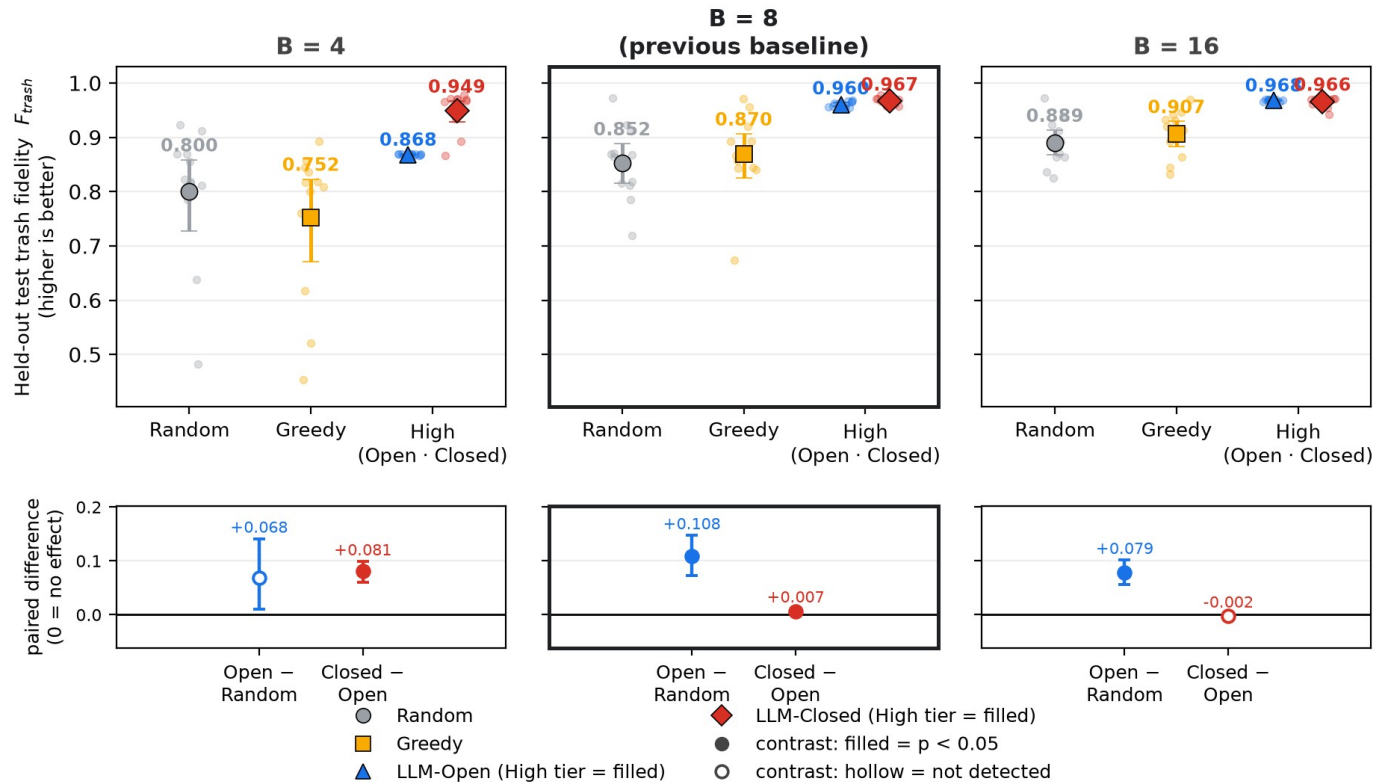
Random and Greedy do not depend on the language model, so each is one result per condition; Low and High each hold an Open and a Closed result.

4 • BUDGET SLICE

Budget changes the value of feedback

B = 4 → 8 → 16 with 4 qubits, Ising chain and High tier fixed; B = 8 (thick frame) is the previous baseline

fixed: 4 qubits • Ising • High model



Semantic proposals stay above non-semantic search at every budget.

Open - Random is positive at all three budgets; at B = 4 it is not detected (+0.068 (8/12, $p = 0.064$)).

The Closed - Open advantage shrinks as the budget grows:

+0.081 at B = 4 → +0.007 at B = 8 → -0.002 at B = 16 (not detected).

Free redesign pays most when the open batch is thinnest; a wide open batch buys the same breadth without feedback.

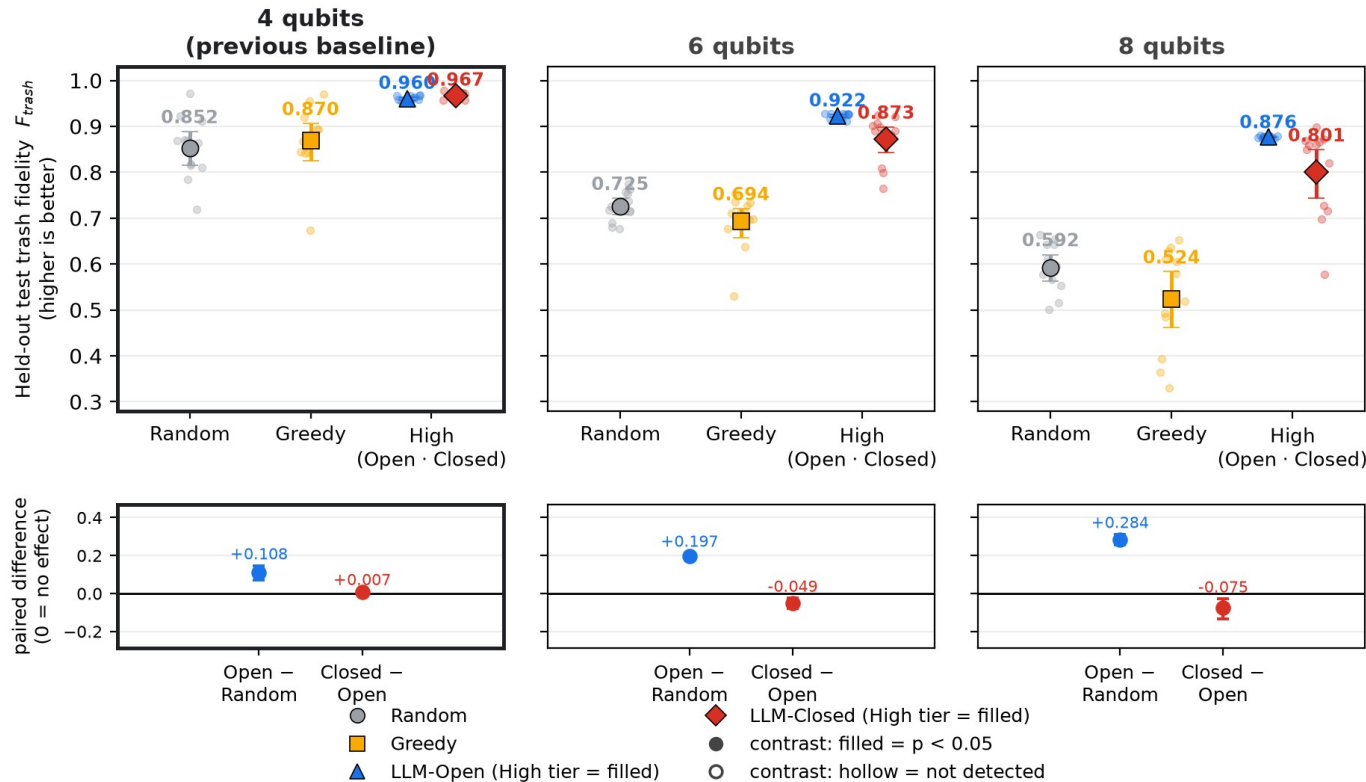
Next: budget changes the amount of search; next we ask whether register width changes the conclusion.

Same trainer, prompts, seeds and selection rule at every budget; adaptive methods always split B as B/2 exploration + B/2 refinement (2+2, 4+4, 8+8).

The semantic advantage grows as the register widens

4 → 6 → 8 qubits with $B = 8$, Ising chain and High tier fixed; 4 qubits (thick frame) is the previous baseline

fixed: $B = 8$ · Ising · High model



1 Absolute fidelity falls as the task widens

Random 0.85 → 0.59; LLM-Open 0.96 → 0.88.

2 Open stays strong; Closed falls below Open

Open - Random +0.108 → +0.197 → +0.284, significant at every width.

Closed - Open +0.007 → -0.049 → -0.075, reversed at 6 and 8 qubits.

Caveat: closed-loop proposals meeting the gate-count contract (the required 3 rotations + 1 CNOT per qubit) fell 92% → 70% → 77%; invalid proposals become random draws.

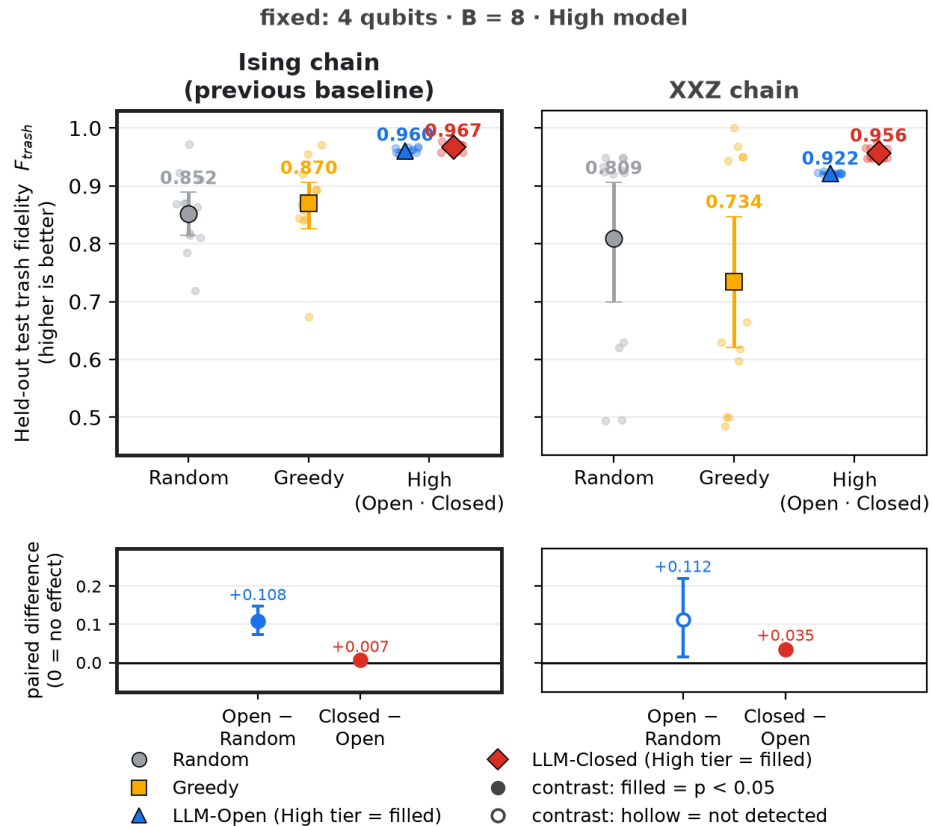
Tested for the Ising chain and the High tier only.

Next: width changes task difficulty; next we change the physical state family at the baseline width.

Same $B = 8$, prompts (qubit count substituted), trainer and seeds at every width; latent and trash registers are always half and half.

The broad ordering survives at the tested XXZ anchor

Ising → XXZ with 4 qubits, B = 8 and High tier fixed; the Ising chain (thick frame) is the previous baseline



The same broad semantic advantage

Both LLM methods have higher means than Random on XXZ (Open 0.92, Closed 0.96 vs Random 0.81).

Per seed: Closed – Random +0.147 (11/12, $p < 0.001$); Open – Random +0.112 (5/12, $p = 0.791$) — Random's seeds spread widely, so that mean gain is not detected per seed.

Closed exceeds Open at this anchor

Closed – Open +0.035 (12/12, $p < 0.001$), larger than at the Ising baseline.

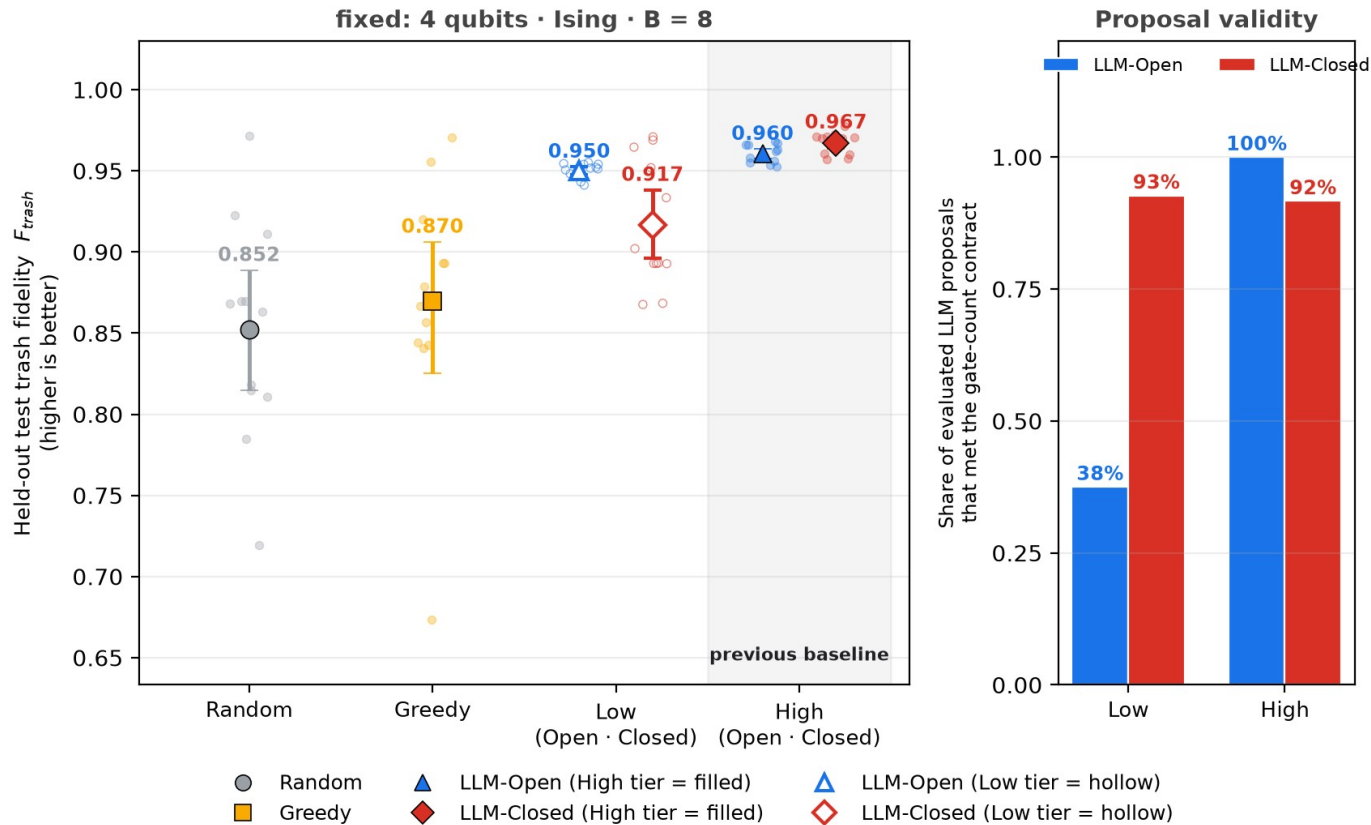
One XXZ point is not the XXZ landscape: budget, width and model tier were not varied for XXZ.

Next: the physics anchor gives one second-family result; the final controlled change is the model itself.

Only the stated Hamiltonian facts change in the prompt (name, formula, parameter range, interaction type); circuit space, trainer and seeds are identical.

Model tier changes reliability more than the Open advantage

Low → High with 4 qubits, Ising chain and B = 8 fixed; High (shaded band) is the previous baseline; groups Random → Greedy → Low → High



The Open advantage is similar for both tiers

Open – Random +0.098 (11/12, $p < 0.001$) for Low, +0.108 (11/12, $p < 0.001$) for High.

The Closed – Open relationship differs

-0.034 (4/12, $p = 0.052$) for Low vs +0.007 (11/12, $p = 0.012$) for High — the baseline's small closed-loop gain is not present for Low.

Proposal validity is a separate failure mode

Only 38% of Low's open-batch proposals met the gate-count contract (Low closed 93%; High 100% / 92%). Invalid proposals become random draws, so validity and architecture quality are entangled in the score.

Low and High are fixed tier labels, not a ranking by score.

Next: the four slices can now be combined to separate robust effects from fragile ones.

Identical prompt bytes, temperature, retry policy and token limit for both tiers; only the model name changes. Random and Greedy are the same runs in both groups.

What survived across the tested slices?

Closed – Open sorted by what the paired test detected; the Open – Random advantage kept its sign in every slice

Closed beats Open

Baseline: B = 8 +0.007 11/12 seeds

B = 4 +0.081 11/12 seeds

XXZ anchor +0.035 12/12 seeds

small budget, the XXZ anchor, and the baseline (slight)

No difference detected

B = 16 -0.002 9/12 seeds

Low tier -0.034 4/12 seeds

a wide open batch, or the Low tier

Open beats Closed

6 qubits -0.049 0/12 seeds

8 qubits -0.075 1/12 seeds

wider registers; closed-loop validity 92% → 70%, 77%

Across the tested one-factor slices, the Open – Random advantage is the robust finding: positive in all 7 tested conditions, significant in 5. The closed-loop advantage is budget-, width-, Hamiltonian- and model-dependent — not a general result.

Proposal validity can confound method differences

An invalid proposal is replaced by a random draw. Closed-loop validity fell to 70% at 6 qubits; Low's open batch was only 38% valid. Part of what looks like a method effect is a generation-validity effect.

Limitations

no interactions · one XXZ anchor · Low only at the baseline condition · 12 paired seeds · noiseless simulation · qubit count and circuit size scale together · not tested: everything else

Next: ranking methods at a fixed budget is the wrong question for practice — how much budget is actually required?

Next: how much budget is actually required?

Three statements from the tested slices, then a targeted boundary-finding experiment rather than a factorial sweep

Robust

Physics-informed Open proposals beat non-semantic search across the tested one-factor slices.

Fragile

The closed-loop advantage is not general: it changes with budget, width, Hamiltonian and model tier.

Operational bottleneck

Valid circuit generation must be separated from architecture quality.

Why: every comparison so far ranks methods at a fixed B; what decides practical use is the budget needed to reach the required accuracy.

Next experiment: the minimum budget for a target fidelity

Before running, fix a target validation fidelity F_{target} (candidates 0.95 and 0.99). B_{min} = the smallest budget reaching F_{target} in at least 10 of the 12 paired seeds — chosen on validation; the held-out test is read only after B_{min} is selected.

API-minimising procedure

- 1 Reanalyse the existing logs first — no new calls.
- 2 Build best-so-far validation curves versus candidate count from those logs.
- 3 Check whether each target is already reached by $B = 4, 8$ or 16 .
- 4 Keep only the conditions whose B_{min} is unresolved.
- 5 Run new LLM calls only at the nearest bracketing budget for those.
- 6 Preserve the same 12 paired seeds for any confirmatory run.
- 7 Report an interval for B_{min} where the exact minimum would cost too much.

Not a blind full sweep — targeted boundary-finding, not factorial completion.